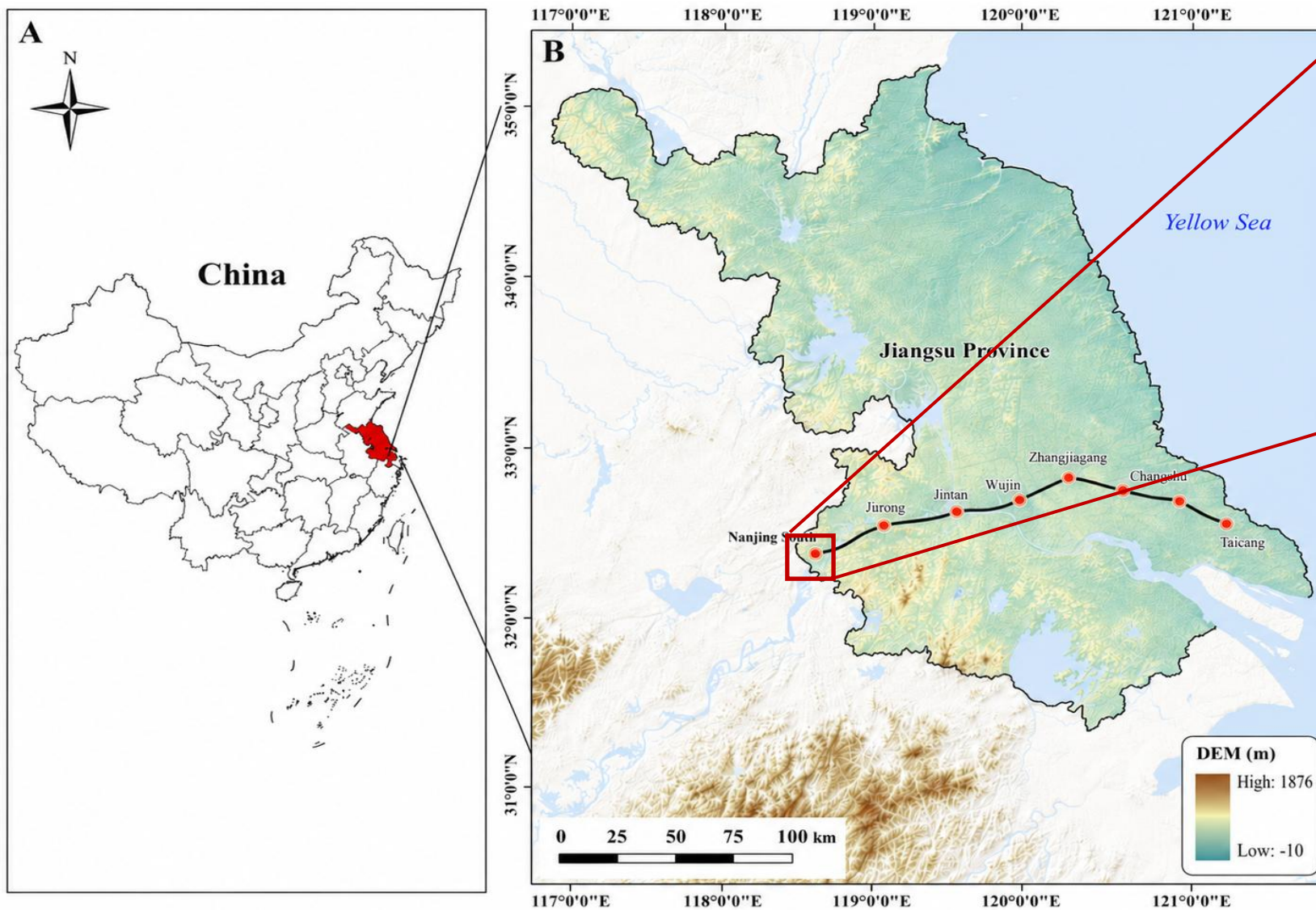
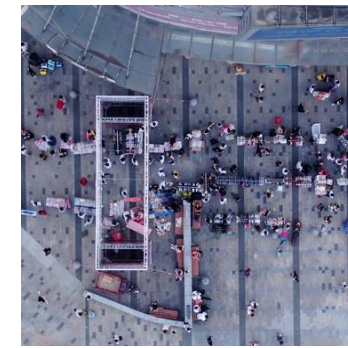


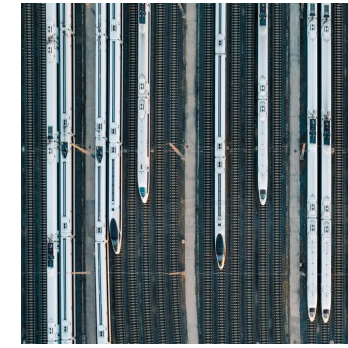


Geographical Location of the Nanjing South-Taicang High-Speed Railway

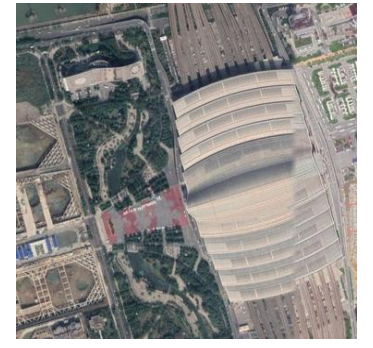
Multi-factor Coupling Effects in High-Speed Railways



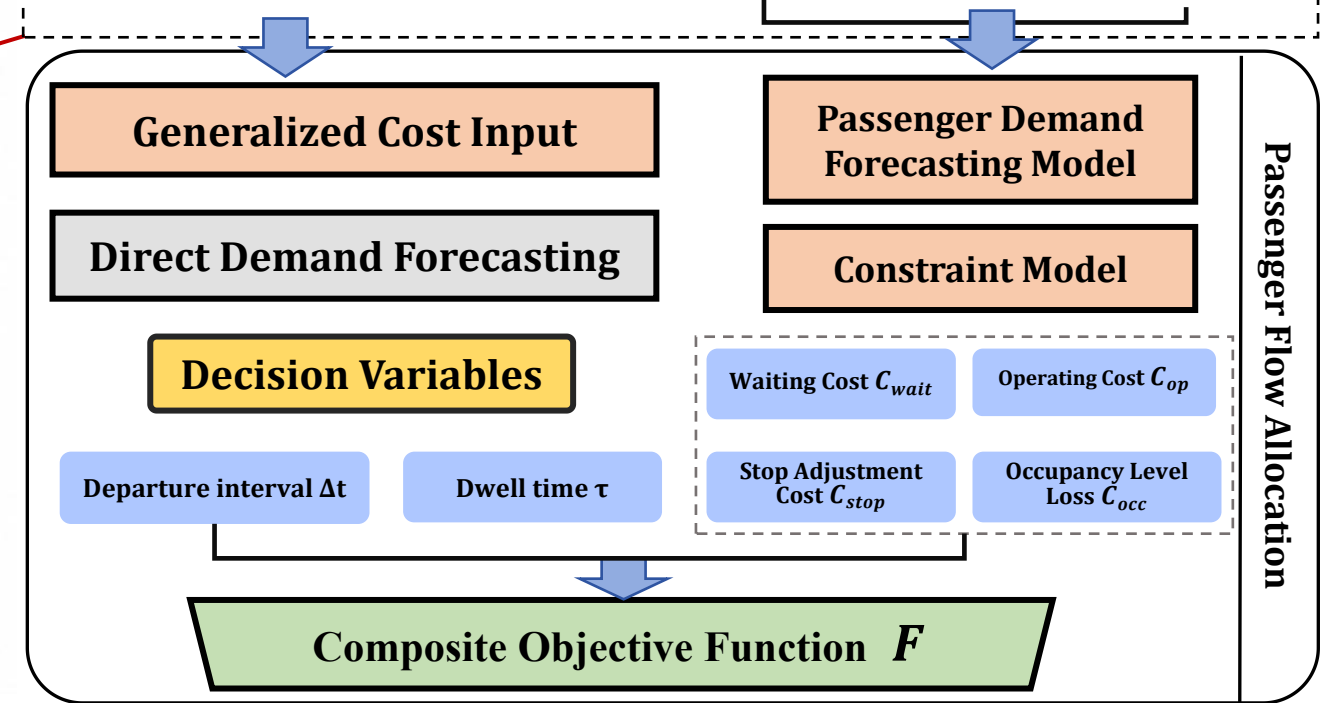
(c) Passenger Demand



(b) Stop-Skipping Decision



(a) Station Service Capacity



Output

$$\Delta t_k^*$$

$$n_k^*$$

$$\tau_{k,j}^*$$

Four search operators with functional roles

$$z_i^{(1)}(g) z_i^{(2)}(g) z_i^{(3)}(g) z_i^{(4)}(g)$$

$$p_r(g) = \frac{(h_r(g) + \varepsilon)^\alpha (w_r(g) + \varepsilon)^\beta}{\sum_{m=1}^4 (h_m(g) + \varepsilon)^\alpha (w_m(g) + \varepsilon)^\beta}$$

Success-history Credit Evaluation

$$h_r(g) = (1 - \rho)h_r(g-1) + \rho h_{min}$$

Scout bees (directed restart)

Adaptive Iterative Search and Optimization Process

Employed bees (neighborhood search)

Population Diversity Evaluation

$$D(g) = \frac{1}{SN} \sum_{i=1}^{SN} \frac{\|x_i^g - \bar{x}^g\|_2}{\|U - L\|_2 + \varepsilon}$$

$$R(g) = \frac{f_{best}(g-1) - f_{best}(g)}{\max\{|f_{best}(g-1)|, \varepsilon\}}$$

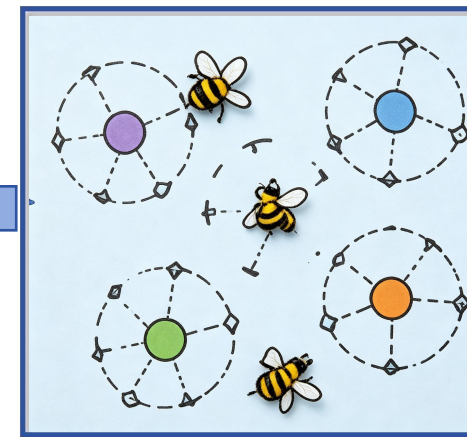
Best-solution Improvement Evaluation

$$G(g) = \begin{cases} 0, & R(g) > \tau, \\ G(g-1) + 1, & R(g) \leq \tau, \end{cases}$$

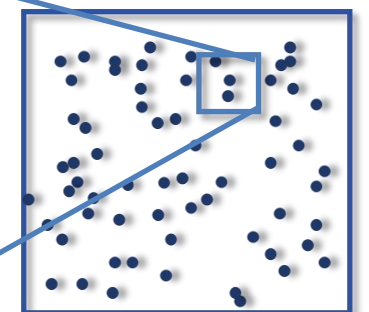
Search Stagnation Detection

State Decision

SHDMS Block



Global search mechanism of population



Initialize the food source population using random-key encoding

Input and Initialize the food source population